

Business and Data Understanding Report

ICU Patient Care Analysis Project

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1. Introduction

Regional Medical Center (RMC) is a 400-bed academic medical center with a 36-bed intensive care unit (ICU) serving a diverse metropolitan population. RMC has observed variation in ICU patient outcomes and resource use and wants a more systematic, data-driven understanding of the patient characteristics, clinical factors, and care patterns associated with these differences (Burton, n.d.). This project uses deidentified electronic health record data to identify patterns that may support clinical practice, quality improvement, and operational decision-making.

This report documents the Business Understanding and Data Understanding phases of the ICU patient care analysis project following the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. It defines the business problem, objectives, stakeholders, responsibilities, analytical goals, and project plan, then describes the Medical Information Mart for Intensive Care (MIMIC)-IV Demo data, summarizes initial exploratory findings, and identifies data-quality considerations that will shape later data preparation and analysis.

2. Business Understanding

2.1 Business Problem and Objectives

RMC lacks a systematic understanding of the factors associated with variation in ICU patient outcomes, care delivery, and resource utilization. Clinical experience is essential for individual care, but analysis across patients and encounters can reveal broader patterns that are difficult to identify from individual cases alone (Burton, n.d.). A data-driven approach adds value by allowing RMC to systematically examine patterns across patients, admissions, ICU stays, treatments, and outcomes that may not be apparent from individual clinical cases. These findings can provide an empirical basis for identifying areas that warrant further clinical or operational investigation.

The primary business objective is to improve RMC's understanding of ICU outcomes and care delivery so that clinical practice, quality improvement, and operational decisions can be supported by evidence. Supporting objectives are to:

- Identify patient and clinical characteristics associated with outcomes such as ICU length of stay, hospital length of stay, mortality, and discharge disposition.
- Describe treatment and care patterns and examine how they vary across patients and outcomes.

- Characterize resource utilization, including length of stay, procedures, medication-related records, and transfers.
- Use the evidence generated to identify areas that may warrant further clinical or operational investigation.

2.2 Analytical Objectives

Table 1. *Analytical Objectives, Focus Areas, and Key Measures.*

Objective	Analytical focus	Key measures / questions
Patient outcomes	Examine associations between patient characteristics, clinical factors, and outcomes.	Age, gender, diagnoses, ICU/hospital Length of Stay (LOS), mortality, discharge disposition.
Treatment and care patterns	Describe medications, procedures, interventions, and monitoring patterns and relate them to selected outcomes.	Medication records, hospital/ICU procedures, inputs, outputs, LOS and mortality.
Resource utilization	Identify variation in the duration and intensity of care and characteristics associated with higher utilization.	ICU/hospital LOS, procedure counts, prescription counts, transfers, ICU interventions.

Table 1 summarizes the analytical objectives of the project and links each objective to its analytical focus and the key measures that will be examined. The current project is primarily descriptive and exploratory rather than predictive. Descriptive statistics and visualizations will be used to characterize the patient population, outcomes, treatment patterns, and resource utilization, while selected statistical methods may be used to examine relationships among variables. Predictive classification or regression will only be considered in later phases if the available sample size, outcome frequency, and data quality support reliable estimation. Relationships identified in these data will be treated as associations rather than causal effects, and analytical questions may be refined as data-quality and sample-size limitations become clearer.

2.3 Stakeholders and Success Criteria

The ICU Medical Director is the primary clinical stakeholder and is interested in evidence that can improve understanding of outcomes, care practices, and potential quality-improvement opportunities. Hospital Administration is primarily interested in operational performance, resource utilization, and quality metrics. Findings, therefore, need to be methodologically sound,

clinically interpretable, and useful for operational planning without overstating what the data can establish.

Project success will be evaluated by whether the analysis addresses the three defined analytical objectives: patient outcomes, treatment and care patterns, and resource utilization. At minimum, the analysis will characterize all four primary outcome measures (ICU LOS, hospital LOS, in-hospital mortality, and discharge disposition), summarize selected treatment and utilization measures, and evaluate at least three relationships between patient or clinical characteristics and these outcomes using descriptive or statistical methods appropriate to the available data. Where inferential methods are used, effect estimates, uncertainty, and statistical significance will be reported as appropriate rather than treating statistical significance alone as evidence of practical importance.

Findings will be considered analytically successful when they are supported by clearly defined measures and appropriate denominators, interpreted within the healthcare context, and accompanied by relevant limitations and potential sources of bias. Given the exploratory nature and 100-patient size of the MIMIC-IV Demo, success will not be defined by achieving a predetermined predictive-performance threshold or statistically significant result. Instead, emphasis will be placed on methodological validity, interpretability, and identification of evidence-supported areas for further investigation. The primary deliverable is a healthcare analytics report suitable for review by hospital leadership and clinical staff.

2.4 Assessment of the Situation and Responsibility

The primary resource is the MIMIC-IV Clinical Database Demo, which contains deidentified hospital and ICU records for 100 patients. MIMIC-IV documentation and data dictionaries support interpretation of table structures, identifiers, coded variables, and deidentification. Python and common data-analysis and visualization libraries are used for preparation, exploration, statistical analysis, and visualization.

Important constraints include the small 100-patient sample, sparse observations in some tables or subgroups, missing data, the absence of free-text clinical notes, data from a single medical center, and the observational nature of the dataset. These limitations restrict generalizability and may prevent reliable subgroup analysis or complex modeling. If a proposed question is not adequately supported, the analysis will be narrowed rather than producing unsupported conclusions.

Because the project uses healthcare data, privacy and responsible interpretation are central. The dataset is deidentified and will be used only for the intended academic analysis; no attempt will be made to re-identify patients. Potential bias in patient representation, clinical practice, and data collection will be considered when comparing groups. Recommendations will be framed cautiously so that exploratory associations are not interpreted as evidence of causation or used to make unsupported conclusions about patient populations.

Potential benefits include providing clinical and administrative stakeholders with evidence to identify patterns in outcomes and resource utilization that warrant further investigation. Potential harms could arise if exploratory associations are interpreted as causal, small subgroup results are generalized beyond the available data, or findings are used to characterize patient groups without adequate consideration of sample size, representation, and clinical context.

2.5 Data Science Goals and Project Plan

The technical goal is to use descriptive and exploratory analysis to characterize ICU outcomes, treatment patterns, and resource utilization, then evaluate selected relationships that can support the business objectives. Given the limited demo sample, the emphasis will remain on interpretable descriptive statistics, visualizations, and appropriately selected statistical methods rather than complex predictive modeling unless the data support it. Table 2 outlines the planned analytical work for the remaining CRISP-DM phases, including data preparation, analysis, evaluation, and communication of findings.

Table 2. *Data Science Project Plan by CRISP-DM Phase.*

CRISP-DM phase	Planned work	Planned completion/time	Milestone/Decision Point
Data Preparation	Select relevant tables and variables; validate subject_id, hadm_id, and stay_id; join tables at the appropriate level; assess missingness; standardize categories; aggregate event-level records to the ICU-stay or admission level; and create derived measures such as hospital LOS and utilization counts.	09/29/2026-10/13/2026	Produce an analysis-ready dataset. Confirm that identifiers, joins, missingness, and selected variables adequately support the planned analyses.
Modeling / Analysis	Examine selected relationships between patient characteristics, diagnoses, treatments, resource utilization, and outcomes using descriptive and statistical methods	10/14/2026-10/26/2026	Complete the primary analyses for patient outcomes, treatment/care patterns, and resource

CRISP-DM phase	Planned work	Planned completion/time	Milestone/Decision Point
	appropriate to sample size and variable type.		utilization. Narrow or modify analyses that are not adequately supported by the data.
Evaluation	Assess statistical and practical relevance, robustness, data limitations, and alignment with clinical and operational objectives. Review whether findings adequately address the analytical objectives without overstating conclusions.	10/27/2026-11/16/2026	Determine which findings are sufficiently supported for inclusion in the final report and identify unresolved limitations or areas requiring further investigation.
Deployment / Communication	Translate findings into a concise report for clinical and administrative stakeholders, including visualizations, interpretations, limitations, and areas for further investigation.	11/17/2026-11/30/2026	Complete and review the final healthcare analytics report for submission.

The project plan accounts for several risks identified during Data Understanding. The small patient sample and low frequency of some outcomes may prevent reliable subgroup comparisons or complex modeling; in those cases, the analysis will emphasize descriptive statistics and appropriately limited comparisons. Sparse categorical groups will be combined only when analytically and ethically justified or excluded from subgroup analysis when estimates would be unreliable. Event-level tables will be aggregated to the appropriate admission or ICU-stay level before comparisons to avoid treating repeated observations as independent cases. If missingness, linkage problems, or insufficient observations prevent a planned analysis from being supported, the analytical scope will be narrowed rather than producing unstable or unsupported conclusions.

3. Data Understanding

3.1 Data Source, Organization, and Inventory

The project uses the MIMIC-IV Clinical Database Demo, an open-access subset of the MIMIC-IV database derived from electronic health records collected during routine clinical care at Beth Israel Deaconess Medical Center. Patient information was captured through hospital electronic health record systems during admission, treatment, and monitoring and was subsequently extracted and processed by the MIT Laboratory for Computational Physiology as part of the MIMIC project. The demo dataset was created by randomly selecting 100 patients from the full database while preserving their associated records, data relationships, and temporal patterns. No additional sampling or filtering was applied beyond patient selection. The course materials do not specify the clinical data-collection timeframe. Files were provided through the course materials in compressed CSV format, and official MIMIC-IV documentation and data dictionaries are used to interpret the data.

The data are organized primarily into HOSP and ICU modules. HOSP contains patient demographics, admissions, diagnoses, laboratory measurements, prescriptions, procedures, and transfers. ICU contains ICU stays and more granular clinical events such as charted measurements, inputs, outputs, and procedures. Records are linked through `subject_id` (patient), `hadm_id` (hospital admission), and `stay_id` (ICU stay). Table 3 provides an inventory of the primary HOSP and ICU tables used in the analysis, including the number of records in each table and its relevance to the project objectives. The MIMIC-IV Demo is publicly available under the Open Data Commons Open Database License v1.0.

Table 3. *Data Inventory and Relevance to the Analysis.*

Module / table	Records	Primary relevance
HOSP: patients	100	Patient demographics and patient-level linkage
HOSP: admissions	275	Hospital encounters, LOS, discharge disposition, mortality
HOSP: diagnoses_icd	4,506	Diagnoses and clinical conditions
HOSP: labevents	107,727	Laboratory measurements
HOSP: prescriptions	18,087	Medication-related treatment patterns
HOSP: procedures_icd	722	Hospital procedures and utilization
HOSP: transfers	1,190	Movement between hospital locations
ICU: icustays	140	ICU encounters and ICU LOS
ICU: chartevents	668,862	Vital signs and other charted observations
ICU: inputevents	20,404	Fluids, medications, and other ICU inputs
ICU: outputevents	9,362	Recorded patient outputs
ICU: procedureevents	1,468	ICU procedures and interventions

3.2 Data Description and Unit of Analysis

MIMIC-IV contains information at several levels: patient → hospital admission → ICU stay → clinical event. A patient may have multiple hospital admissions, an admission may include an ICU stay, and each stay or admission may contain many event-level records. The ICU stay is the primary unit of analysis for ICU-focused questions, while hospital-admission-level measures are used when the outcome or utilization measure is defined at the hospitalization level. The descriptive hospital-level results in this report summarize the 275 hospital admissions represented in the demo data, whereas ICU-specific results are based on the 140 ICU stays. For subsequent ICU-focused analyses, hospital-level measures will be linked to the corresponding ICU population when required by the analytical question. Event-level data will be aggregated before comparisons when necessary to avoid treating repeated records as independent patients.

The course-provided documentation does not specify the clinical collection timeframe represented by the supplied demo files. In addition, temporal fields are deidentified, limiting the interpretation of calendar-time patterns and trends. Temporal variables are therefore used primarily to calculate and characterize within-record measures, such as hospital and ICU length of stay, and to preserve the sequence of events within an encounter rather than to evaluate population-level trends over calendar time. Table 4 presents the primary variables used in the

analysis. Variable definitions and descriptions were informed by the official MIMIC-IV documentation (MIT Laboratory for Computational Physiology, 2023).

Table 4. *Key Variables and Definitions Used in the Analysis.*

Variable	Data Type	Level	Observed Values / Range	Role / Interpretation
anchor_age	Numeric	Patient	21–91 years	Patient age
gender	Categorical	Patient	M, F	Patient demographic characteristic
race	Categorical	Admission	Multiple recorded categories	Recorded race for the hospital admission
admission_type	Categorical	Admission	Multiple admission categories	Type of hospital admission
hospital_expire_flag	Binary	Admission	0, 1	Indicates in-hospital mortality
discharge_location	Categorical	Admission	Multiple discharge destinations	Hospital discharge disposition
admittime, dischtime	Date/time	Admission	Deidentified timestamps	Used to calculate hospital LOS
los	Numeric	ICU stay	0.02-20.53 days	ICU LOS in days
icd_code	Categorical/code	Admission	ICD diagnosis/procedure codes	Diagnosis or procedure classification
drug	Text/categorical	Admission	Medication names	Prescription-related treatment patterns
itemid	Identifier/code	Event	Item identifiers	Identifies charted measurements, inputs, outputs, or ICU procedures
valuenum	Numeric	Event	Varies by measurement	Numeric value for applicable clinical measurements

Initial descriptive statistics show a mean patient age of 61.75 years (median 63; range 21–91). Across 140 ICU stays, mean ICU LOS was 3.68 days, and median LOS was 2.16 days. Across 275 hospital admissions, mean hospital LOS was 6.88 days, and median LOS was 4.85 days. These differences between means and medians suggested right-skewed LOS distributions and motivated closer exploration.

3.3 Data Exploration Report

Exploratory analysis used Pandas for data manipulation and descriptive statistics and Matplotlib/Seaborn for visualization. The analysis focused on the project's three analytical objectives and maintained the appropriate denominator for patient-, admission-, ICU-stay-, and event-level findings. Given the temporal limitations described in Section 3.2, the initial exploration focuses on patient, outcome, treatment, and resource-utilization patterns rather than calendar-time trends.

3.3.1 Patient Population and Clinical Characteristics

The 100-patient sample was primarily composed of middle-aged and older adults. Ages ranged from 21 to 91 years, with a mean of 61.75 years and a median of 63 years. As illustrated in Figure 1, the age distribution was concentrated around the 60–70-year range. The sample included 57 male and 43 female patients.

Race was recorded at the admission level rather than the patient level. As shown in Figure 2, White was the most frequently recorded race category, accounting for 170 of 275 admissions (61.8%), followed by Black/African American with 48 admissions (17.5%). Several other race categories contained relatively few observations, which may limit reliable subgroup comparisons.

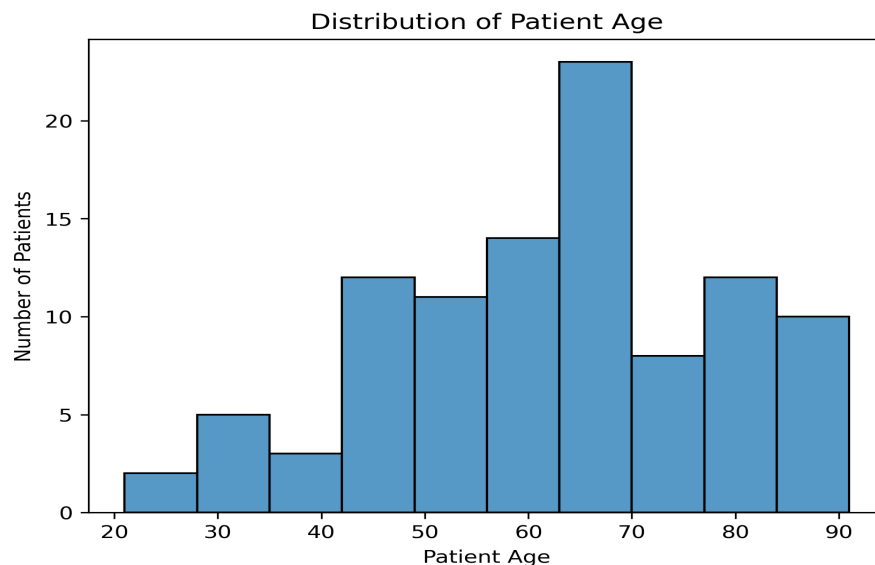


Figure 1. *Distribution of patient age.*

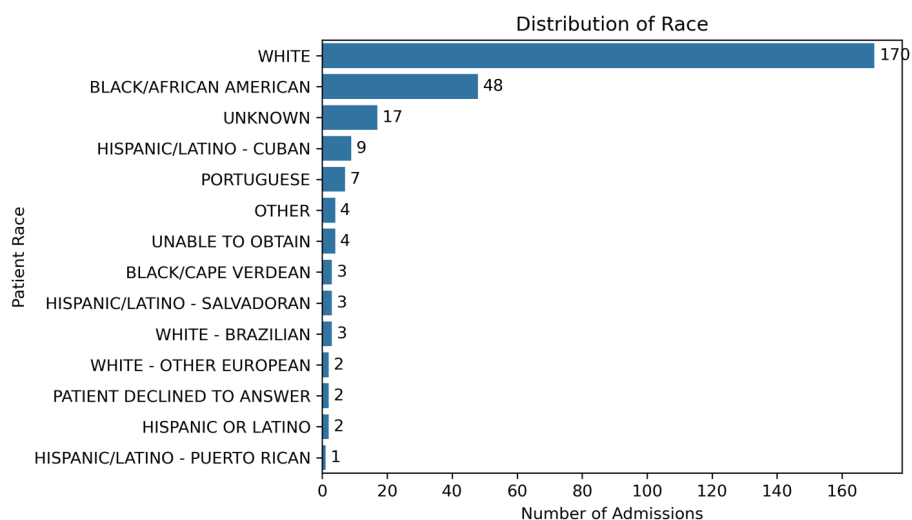


Figure 2. *Recorded race across hospital admissions.*

Hospital admissions were commonly emergency-related or observational. As shown in Figure 3, EW EMER. was the most frequent admission type, accounting for 104 admissions (37.8%), followed by Observation Admit (45), Urgent (38), and EU Observation (30).

Diagnosis records reflected multiple conditions per admission. As illustrated in Figure 4, common diagnoses included hypertension, hyperlipidemia, acute kidney failure, hypothyroidism, obesity, and anemia. Approximately 31.16% of diagnosis descriptions contained “unspecified,” which may limit the specificity of detailed clinical groupings.

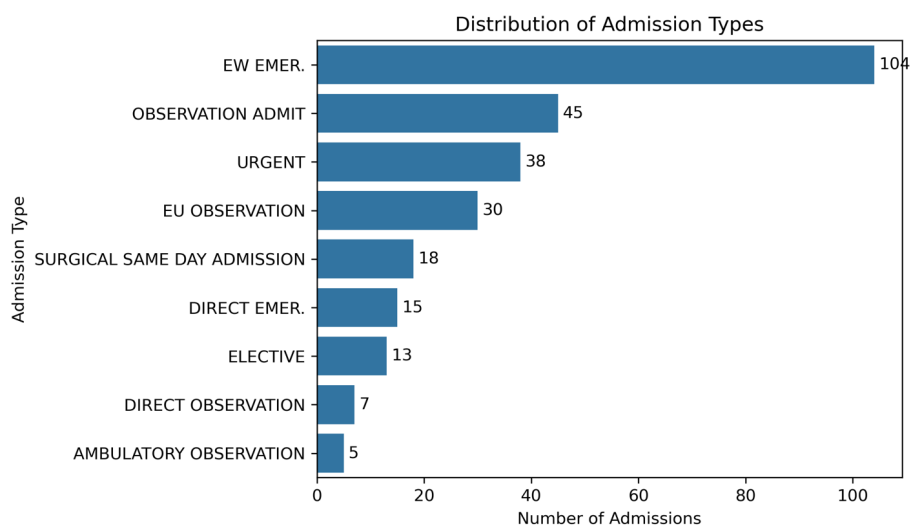


Figure 3. *Distribution of hospital admissions by admission type.*

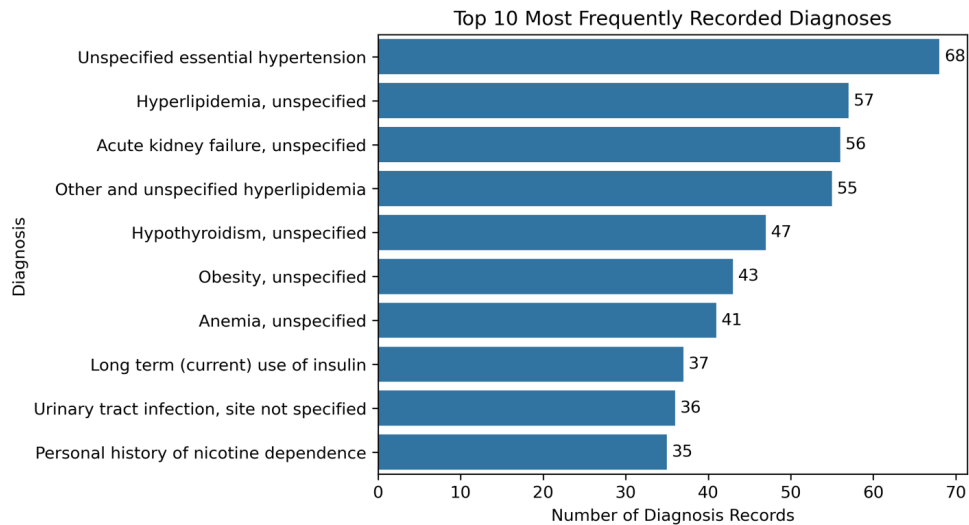


Figure 4. *Ten most frequently recorded diagnoses.*

Because patients may have multiple admissions and diagnoses, these counts should not be interpreted as patient prevalence. Subsequent analyses must preserve the correct denominator: 100 unique patients for patient-level characteristics, 275 admissions for admission-level characteristics, and 4,506 diagnosis records for diagnosis frequencies.

3.3.2 Patient Outcomes

Both ICU and hospital length of stay (LOS) were strongly right skewed. As shown in Figure 5, most ICU stays were relatively short, with a median LOS of 2.16 days and a mean of 3.68 days, while the maximum reached 20.53 days. Hospital admissions showed a similar pattern, with a median LOS of 4.85 days, a mean of 6.88 days, and a maximum of 44.93 days. The difference between the mean and median LOS indicates that a smaller number of prolonged stays increased the average and may represent comparatively high resource utilization.

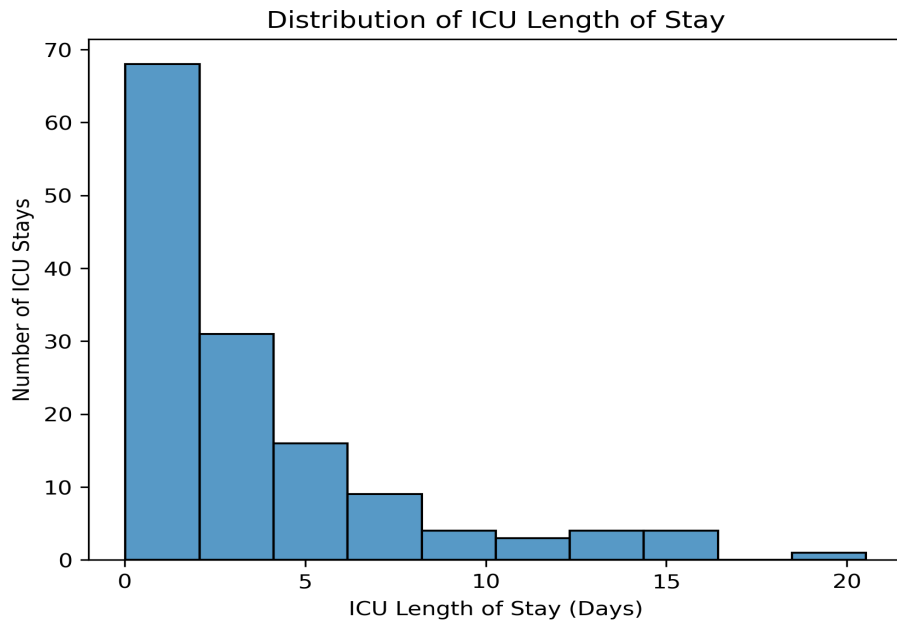


Figure 5. *Distribution of ICU length of stay.*

In-hospital mortality was uncommon. As shown in Figure 6, 15 of 275 admissions (5.45%) resulted in death, while 260 (94.55%) did not. This strong imbalance limits mortality subgroup comparisons and may produce unstable estimates in small groups. Discharge disposition also varied across admissions. As illustrated in Figure 7, home health care (76) and home (72) together accounted for 53.8% of admissions, while other patients were discharged to skilled nursing, rehabilitation, long-term acute care, hospice, or other destinations.

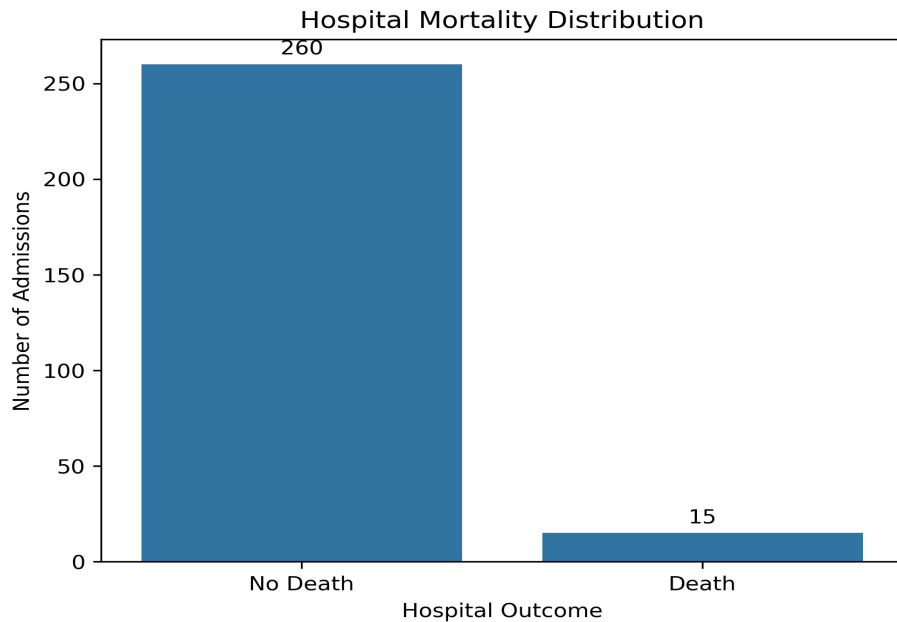


Figure 6. *In-hospital mortality across hospital admissions.*

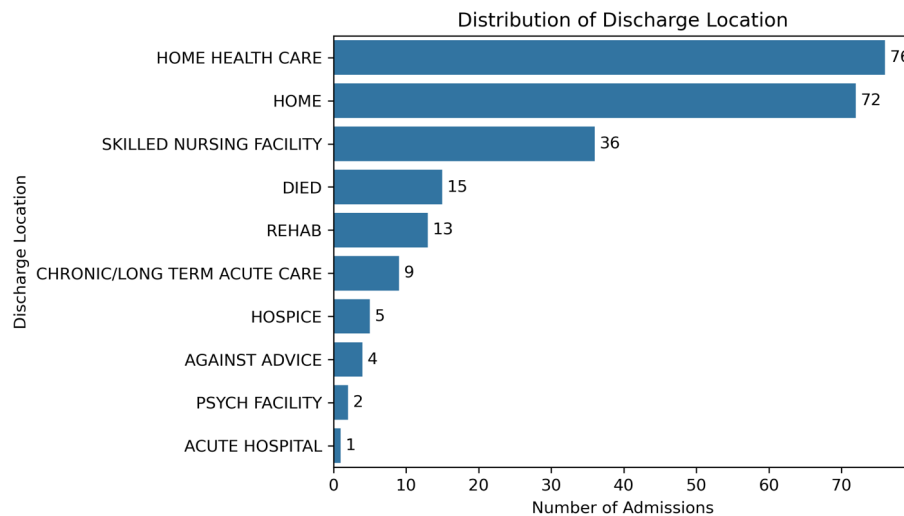


Figure 7. *Discharge disposition across hospital admissions.*

3.3.3 Treatment Patterns and Resource Utilization

Treatment records showed broad variation in care. At the hospital level, insulin was the most frequently prescribed medication, while commonly documented procedures included vascular access, nutritional support, and mechanical ventilation. ICU procedure records frequently reflected vascular access, diagnostic imaging, cardiovascular monitoring, respiratory support, and catheter-related interventions. These frequencies represent clinical records rather than unique patients or stays and therefore require aggregation before outcome comparisons.

Resource utilization varied substantially across admissions. Hospital procedure counts ranged from 0 to 23 per admission (median = 1), and 32.0% of admissions had no ICD procedure records. As shown in Figure 8, medication-related utilization also varied considerably across hospital admissions. Prescription counts ranged from 0 to 348 per admission (median = 48), with 9.09% of admissions having no prescription records. Transfer events ranged from 0 to 8 per admission (median = 1), and 36.36% of admissions had no recorded transfer events. Overall, these distributions were right skewed, indicating that a smaller number of hospitalizations accounted for comparatively high recorded utilization.

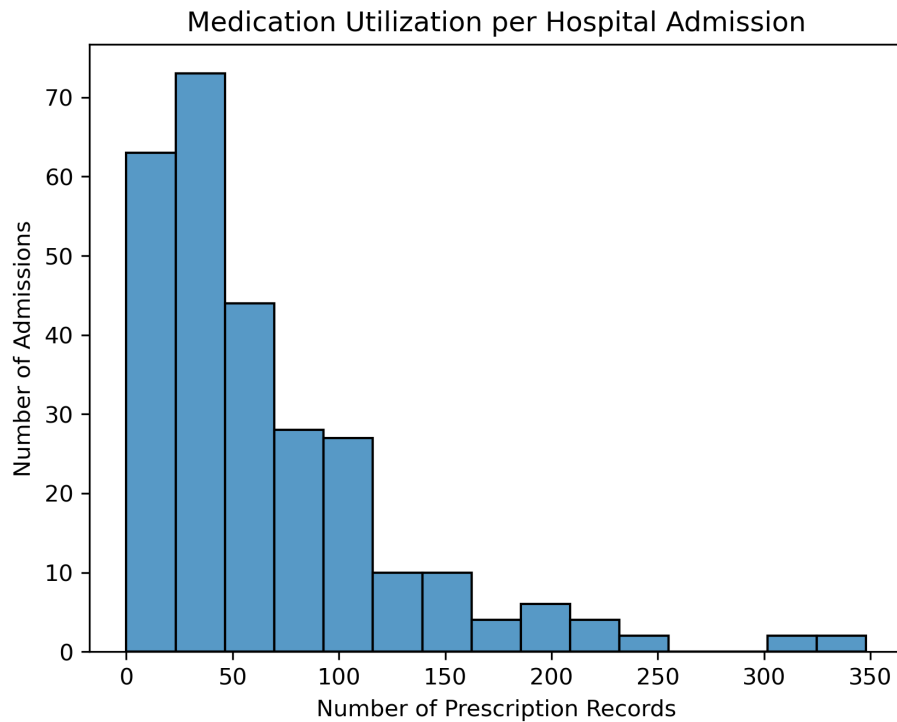


Figure 8. *Medication-related utilization per hospital admission.*

The exploration therefore identifies several priorities for later analysis: prolonged LOS, high-utilization admissions, differences in treatment intensity, and the relationship of these patterns to diagnoses and patient characteristics. Because the data are observational, treatment and utilization patterns may reflect underlying illness severity and should not be interpreted as causing outcomes.

3.3.4 Preliminary Relationships

Initial exploratory comparisons were also conducted to determine whether selected patient characteristics differed between patients with a single ICU stay and those with multiple ICU stays. Of the 100 patients, 74 had one ICU stay and 26 had multiple ICU stays. Mean age was similar between the groups (61.91 years for patients with one ICU stay and 61.31 years for patients with multiple ICU stays), suggesting little descriptive difference in age.

The proportion of patients with multiple ICU stays differed descriptively by gender: 32.6% of female patients had multiple ICU stays compared with 21.1% of male patients. Given the small sample, these results should be interpreted as preliminary descriptive patterns rather than evidence of a statistically established relationship. Further analysis may evaluate whether repeated ICU utilization is associated with patient characteristics, diagnoses, or other indicators

of clinical complexity. Table 5 summarizes preliminary comparisons of age and gender between patients with one ICU stay and those with multiple ICU stays.

Table 5. *Preliminary Comparison of Patient Characteristics by Number of ICU Stays.*

Characteristic	One ICU Stay	Multiple ICU Stays
Patients	74	26
Mean age	61.91	61.31
Median age	63.00	61.50
Female patients — % with one vs. multiple ICU stays	67.4%	32.6%
Male patients — % with one vs. multiple ICU stays	78.9%	21.1%

3.4 Data Quality Assessment

Table 6. *Data Quality Issues, Implications, and Planned Responses*

Issue	Evidence observed	Implication / response
Small sample and rare outcomes	100 patients; 15 in-hospital deaths across 275 admissions	Limit subgroup comparisons and complex modeling; emphasize descriptive/exploratory analysis.
Repeated observations	Patients may have multiple admissions; admissions/stays contain many diagnoses and events	Maintain patient/admission/stay denominators and aggregate event-level records before analysis.
Sparse demographic categories	Several race categories contain few admissions	Avoid unstable subgroup conclusions; combine categories only when analytically and ethically justified.
Inconsistent race records	4 of 100 patients had inconsistent race classifications across admissions	Treat race as admission-level recorded information and document inconsistency.
Diagnosis specificity	31.16% of diagnosis descriptions contain “unspecified”	Use caution when grouping or interpreting diagnoses.
Zero utilization records	32.0% no ICD procedures; 9.09% no prescriptions; 36.36% no transfers	Do not equate absence in one table with absence of all care; verify related tables and definitions.
Deidentification	Dates and age-related fields are deidentified	Interpret temporal variables using MIMIC documentation; focus on within-record intervals where appropriate.

Initial exploration identified several data-quality issues that directly affect the analytical strategy. The most important are limited sample size, repeated observations at multiple levels, sparse

categories, deidentification, diagnostic specificity, and the need to distinguish a missing record from absence of clinical care. Table 6 summarizes the primary data-quality issues identified during exploration and their implications for subsequent data preparation and analysis.

The initial data-quality assessment identified the issues summarized above. During Data Preparation, these checks will be extended through a systematic review of missingness, identifier integrity, linkage validity, and extreme values before analytical datasets are finalized. Expected uniqueness of `subject_id`, `hadm_id`, and `stay_id` will be verified at their appropriate levels; ICU stays will be checked for valid admission links; important analytical variables will be assessed for missingness; and extreme clinical values will be reviewed to distinguish plausible observations from potential data-quality problems.

3.5 Data Understanding Summary and Implications

The MIMIC-IV Demo provides interconnected patient, hospital, ICU, and event-level data that can support the project's three analytical objectives, but the 100-patient sample requires a focused analytical scope. The population is largely middle-aged and older, admissions are frequently emergency-related, and common diagnosis records reflect substantial chronic disease and comorbidity. ICU and hospital LOS are right-skewed, mortality is rare, and treatment and resource-utilization records are unevenly distributed across hospitalizations.

These findings support continued analysis of patient outcomes, treatment/care patterns, and resource utilization, while also indicating that later work should prioritize interpretable methods and carefully selected comparisons. Data preparation will need to preserve the correct unit of analysis, aggregate event-level records, validate identifiers and joins, handle missingness and sparse categories, standardize coded variables, and create derived measures such as utilization counts and LOS where appropriate.

These limitations will directly determine the complexity of subsequent analyses: questions that cannot be supported reliably by the available sample size, outcome frequency, or data completeness will be narrowed or addressed descriptively rather than through unstable statistical or predictive models.

4. Conclusion

The Business Understanding phase established a clear need for evidence-based analysis of ICU outcomes, care patterns, and resource utilization at RMC. The Data Understanding phase showed that the MIMIC-IV Demo contains relevant information across linked hospital and ICU tables and revealed meaningful variation in length of stay, treatment activity, discharge outcomes, and recorded resource use. At the same time, the small sample, rare mortality events, repeated observations, sparse categories, and deidentification constrain the strength and generalizability of conclusions.

The next phase will, therefore, focus on preparing a well-defined analytical dataset at the appropriate level, resolving key data-quality issues, and selecting a limited number of relationships that directly address the analytical objectives. This focused approach will allow subsequent analysis to remain aligned with RMC's clinical and operational goals while avoiding unsupported conclusions from the demo dataset.

5. AI Diligence Statement

In preparing this report, I used [ChatGPT \(OpenAI\)](#) to assist with condensing and reorganizing an earlier draft, refining wording for clarity and conciseness, and improving the structure and presentation of the report. I also used ChatGPT for guidance on document formatting and organization. The underlying data analysis, results, interpretations, and conclusions were developed by me. I reviewed, edited, and verified all AI-assisted content before including it in the final report and take full responsibility for the accuracy, integrity, and presentation of the submitted work.

References

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